

Open QHPC Software Ecosystem (openQSE) Github Repositories & Announcements

openQSE Community
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ORNL is managed by UT-Battelle LLC for the US Department of Energy

Administrative

- **openQSE Day: July 24th**

Time	Item
8:30 - 9:00	Gather
9:00 - 9:30	Georgia Tourassi (ORNL Quantum Strategy)
9:30 - 9:45	Amir Shehata (Where we were, where we're at and where we're going)
9:45 - 12:00	Workstreams - Session 1 • Software Architecture • Software Architecture and Workflows • Quantum Resource Interface • Compiler Infrastructure
12:00 - 1:00	Working Lunch
1:00 - 1:30	Laura Schulz (Keynote)
1:30 - 4:30	Workstreams - Session 2
4:30 - 5:30	Report back and cross workstream discussion
5:30	Adjourn

openQSE Day Working Groups

Working Group	Focus Area
Software Architecture Working Group	Evolves the openQSE architecture as new technical, operational, and scaling requirements emerge. Maintains architectural coherence across all components.
System Architecture and Workflows	Defines how quantum systems integrate into HPC centers. Develops representative hybrid workflows, deployment models, and system-level use cases that position openQSE within the broader HPC–QC ecosystem.
Compiler Infrastructure Working Group	Defines the compiler stack within the openQSE architecture, including IR layers, optimization boundaries, and integration points with runtime and resource interfaces.
Quantum Resource Management and Interfaces Working Group	Develops a unified resource interface model across existing approaches such as QRMI (IBM) and QDMI (MQSS). Explore integration with various resource management systems.

Working Group

Survey to determine working group assignments:

- <https://openqse.atlassian.net/wiki/x/CgC3B>

Per Working Group Goals

- **Milestones:** Define milestones for the next year
- **Working Group Interactions:** Define deliverables to/from other working groups

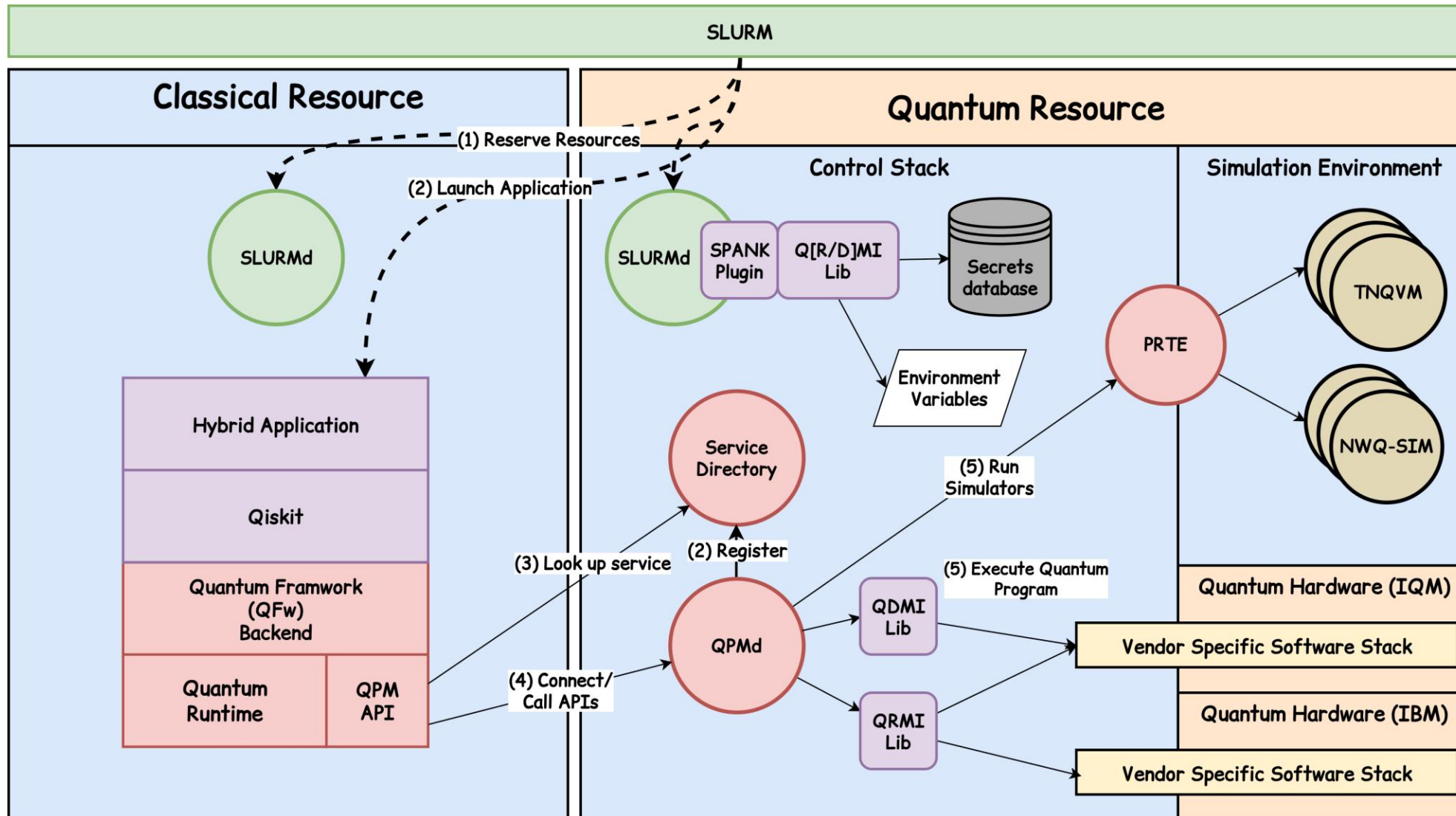
Distributed Execution Framework (DEFw)

- **C Wrapper over Python:** Embeds python in a C Program
- **Communication Infrastructure:** Provides a “standard” communication layer which allows multiple instances of DEFw to communicate together
- **Provides an extensible RPC Layer:** Infrastructure to add services quickly
- **SWIG:** uses swig to allow C function calls from python
- Future work:
 - Integrate with libfabric
 - Improve RPC mechanism
 - Expose as a C library to expand beyond python support
- Show DEFw code structure

The Quantum Framework (QFw)

- **DEFw Usage:** core infrastructure
 - All python applications run within the DEFw
 - Allows applications to discover other DEFw services
- **PRTE:** Uses PRTE for process management
 - Supports multi-simulator runs
- Supports NWQ-SIM and TNQVM

The Quantum Framework



QFw – Future Work

- **Runtime Integration:** AMD/ORNL are working on a runtime to handle quantum workload
- **Resource Management and Interfaces:** IBM/ORNL/MQSC are working on
 - unifying quantum resource interfaces
 - integration with multiple resource managers (SLURM, FLUX, LSF, PBSpro)
- **Incorporate job admission controls:** Rate based, Credit based
- **Incorporate device scheduling:** scheduling algorithms to manage quantum task execution order on device
- **Real Hardware + Simulator:** Allow an environment where concurrent quantum tasks can run on multi-hardware and simulators simultaneously
- **Authentication:** on-prem authentication is different from cloud authentication

High-Level Design

- <https://github.com/openQSE/QFw/blob/main/docs/hld.md>
- Next Step in the software architecture working group
- Detail out the reference architecture published in: <https://arxiv.org/pdf/2604.20912>
- Contributions/Comments welcome

QFw – Demonstration